The R code is:

```
> block1 <- list(number=1,timestamp="May
30 2020 01:15:00",data="Princeton
University",parent=NA)

> block2 <- list(number=2,timestamp="May
30 2020 07:15:00",data="University of
Oxford",parennt=1)

> block3 <-list(number=3,timestamp="May
30 2020 09:45:00",data="CHARUSAT",pare
nnt=2)

> blockchain = list(block1, block2,
block3)

> blockchain

[[1]]
[[1]]$number
[1] 1

[[1]]$timestamp
[1] "May 30 2020 01:15:00"

[[1]]$data
[1] "Princeton University"

[[1]]$parent
[1] NA


[[2]]
[[2]]$number
[1] 2

[[2]]$timestamp
[1] "May 30 2020 07:15:00"

[[2]]$data
[1] "University of Oxford"

[[2]]$parennt
[1] 1

[[3]]
[[3]]$number
[1] 3

[[3]]$timestamp
[1] "May 30 2020 09:45:00"

[[3]]$data
[1] "CHARUSAT"

[[3]]$parennt
[1] 2
```

Validating the block: In R, user defined functions play a role to validate the block in the blockchain.

The R code here is:

```
validateblock = function(blockchain) {
+     if (length(blockchain) >= 2) {
+
+             if (blockchain[[2]]$parent
!= blockchain[[1]]$number) {
+                 return(FALSE)
+             }
+
+     }
+     return(TRUE)
+ }
> validateblock(blockchain )

[1] TRUE
```

Hash function: Each block in a blockchain is identified by a hash value. In R, to generate hash, the digest package is to be used.

The R code is:

```
> library(digest)
> digest("A blockchain is a chain of
blocks", "sha256")

[1] "5c2005976411a1 628fabcdde3ac04b
e563d18943e7 10b7446afa2a8a5fab9abc"
```

Packages for the effective implementation of a blockchain in R

R programming is the right platform to implement a blockchain in a simpler and effective manner. It contains several packages (listed below) through which we can implement a blockchain effectively.

(i) **rockchain:** This provides interfaces to *coinmarketcap.com,* the *blockchain.info* ticker and Bitcoin wallet data.

(ii) **ether:** Ethereum is another renowned cryptocurrency that has some attractive characteristics. It uses JSON-RPC API in order to communicate with the Ethereum network.

(iii) **crypto:** This package delivers the historic OHLC cryptocurrency market data for all coins and exchanges.

(iv) **coinmarketcapr:** This package helps in extracting and monitoring the price and market cap of various cryptocurrencies from CoinMarketCap.

(v) **rbtc:** This package provides the APIs for Bitcoin and utility functions that address the creation and content analysis of a blockchain.

The blockchain has a complex structure. Implementing it seems difficult for novices. In this article, we have provided simple guidelines using R programming, in order to implement it in a simpler way. We have discussed three main functionalities with R code for the implementation. In the first step, we looked at creating individual blocks in a blockchain using the list object of R. In the second step, we have designed a user-defined function to validate the blockchain. In the third step, we have explored how to create a hash function using the digest library of R. In addition to this, we have covered some important packages related to blockchain in R. END

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